

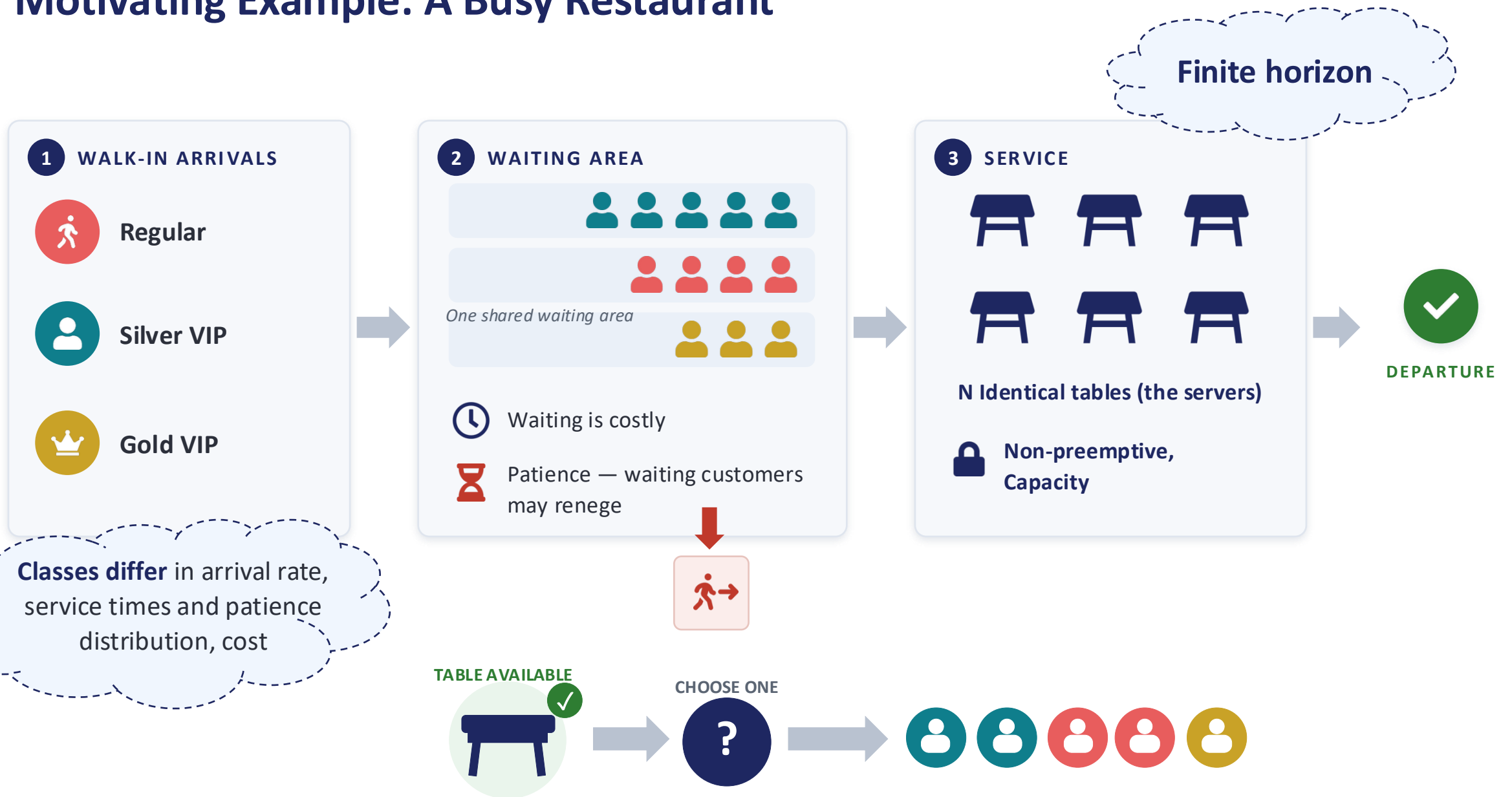
Scheduling a Multi-Class $G/GI/N+GI$ Queue

A Fluid Control Problem and Lagrangian Reformulation

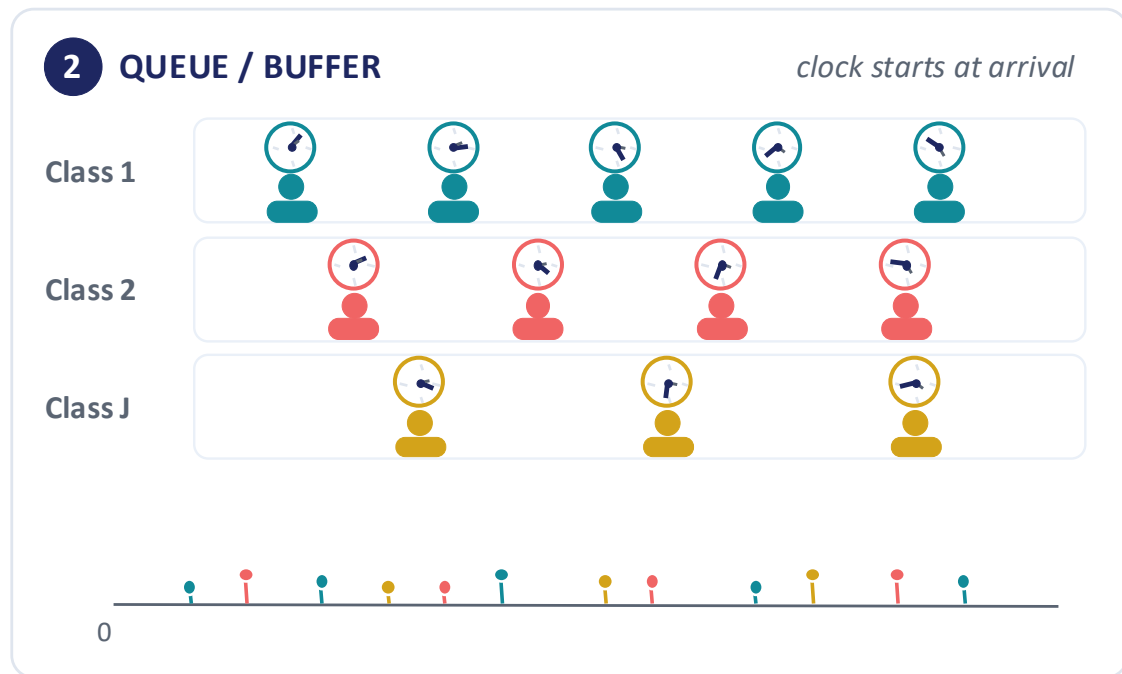
Haoyu Wu · Bahar Taşkesen · Amy Ward

2026 August at Lehigh University

Motivating Example: A Busy Restaurant



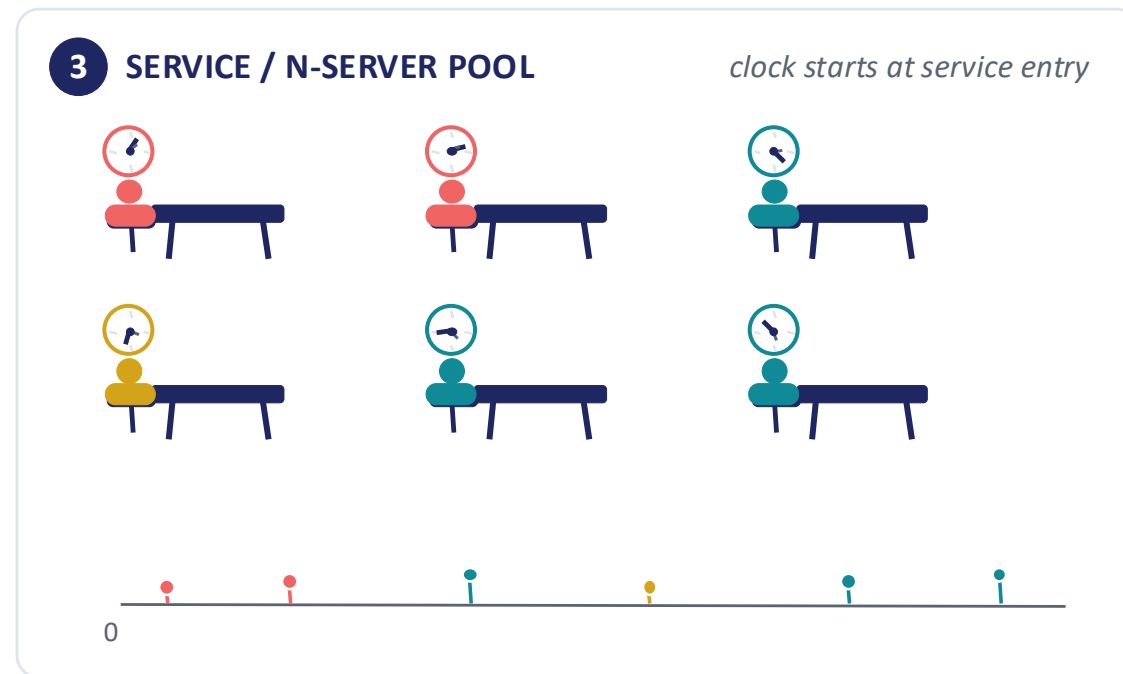
The State of the System: Ages



WAITING-AGE COUNTING MEASURE

$\gamma_j^N(t, \cdot) \in M_+([0, H_j^r])$, $H_j^r = \sup \text{ support of patience}$

$\gamma_j^N(t, A) = \# \{ \text{class-}j \text{ waiting : age} \in A, \text{ Borel } A \subseteq [0, H_j^r] \}$



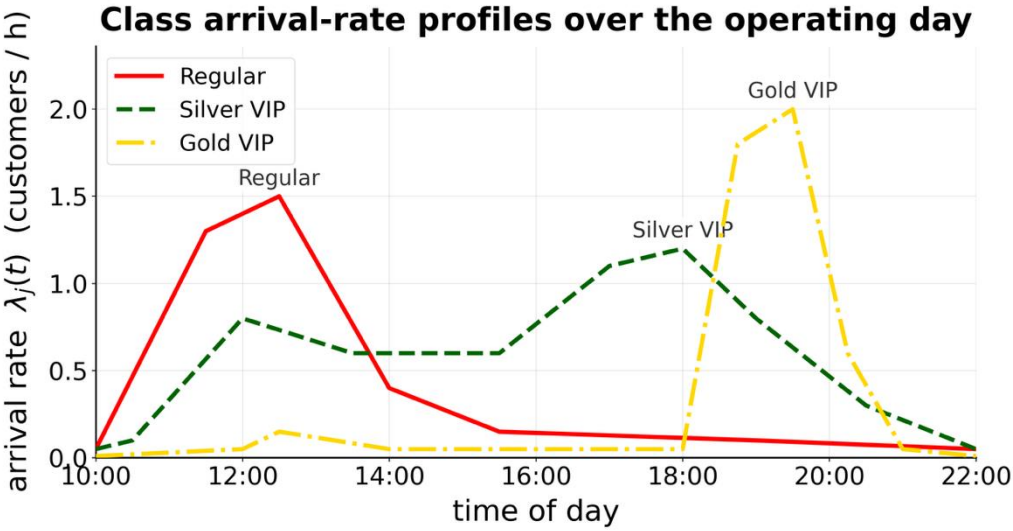
SERVICE-AGE COUNTING MEASURE

$v_j^N(t, \cdot) \in M_+([0, H_j^s])$, $H_j^s = \sup \text{ support of service}$

$v_j^N(t, B) = \# \{ \text{class-}j \text{ in service : age} \in B, \text{ Borel } B \subseteq [0, H_j^s] \}$

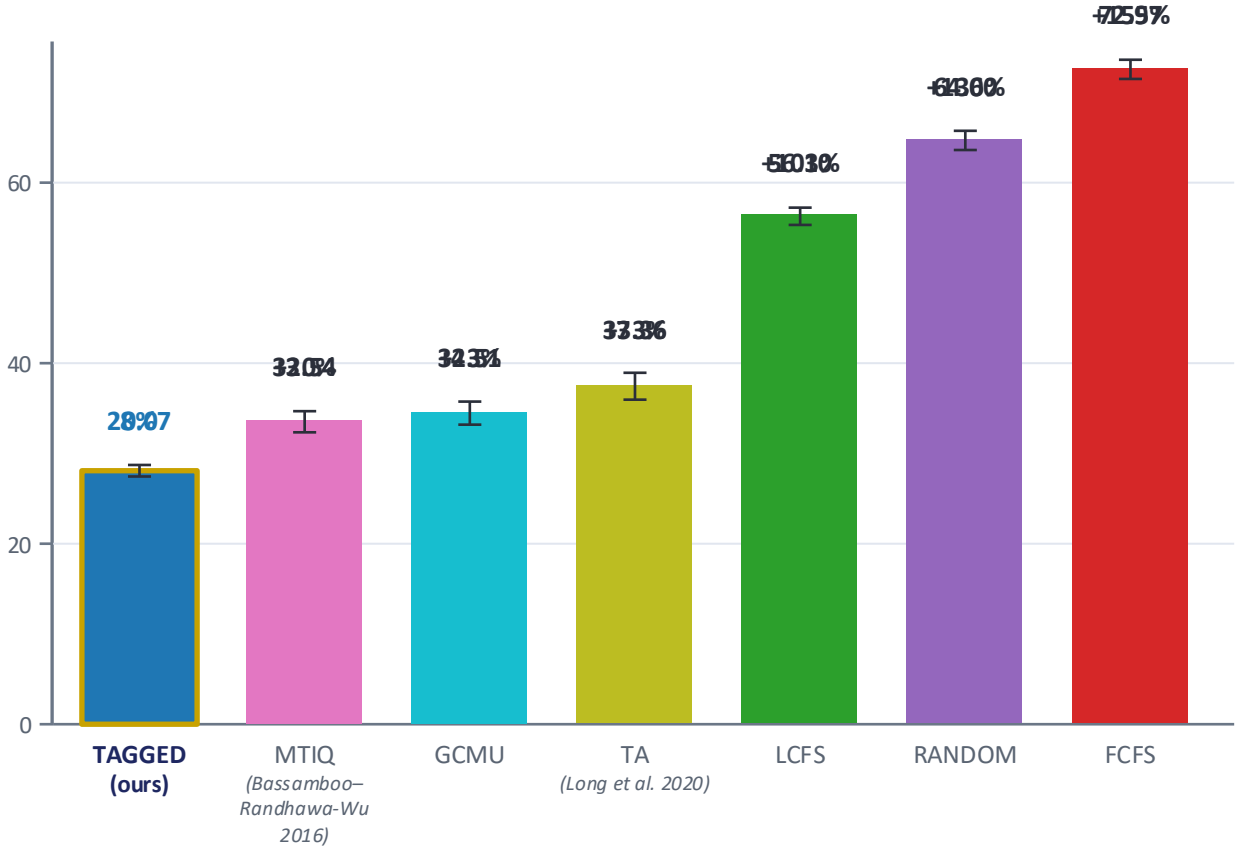
Numerical Results: Three-Class Case

Property	Class 1 — Regulars	Class 2 — Silver VIPs	Class 3 — Gold VIPs
Arrivals	lunch peak (avg 0.41/h)	all day (avg 0.62/h)	dinner wave (avg 0.32/h)
Patience	Medium (~0.67 h)	High (~1.5 h)	Low (~0.5 h)
Service	Fast (~0.67 h)	Slow (~2.0 h)	Medium (~0.83 h)
Abandonment cost	Low (1)	Medium (3)	High (15)
Holding cost	Low ($0.5 \cdot w^2$)	Medium ($1.0 \cdot w^2$)	High ($2.0 \cdot w^2$)



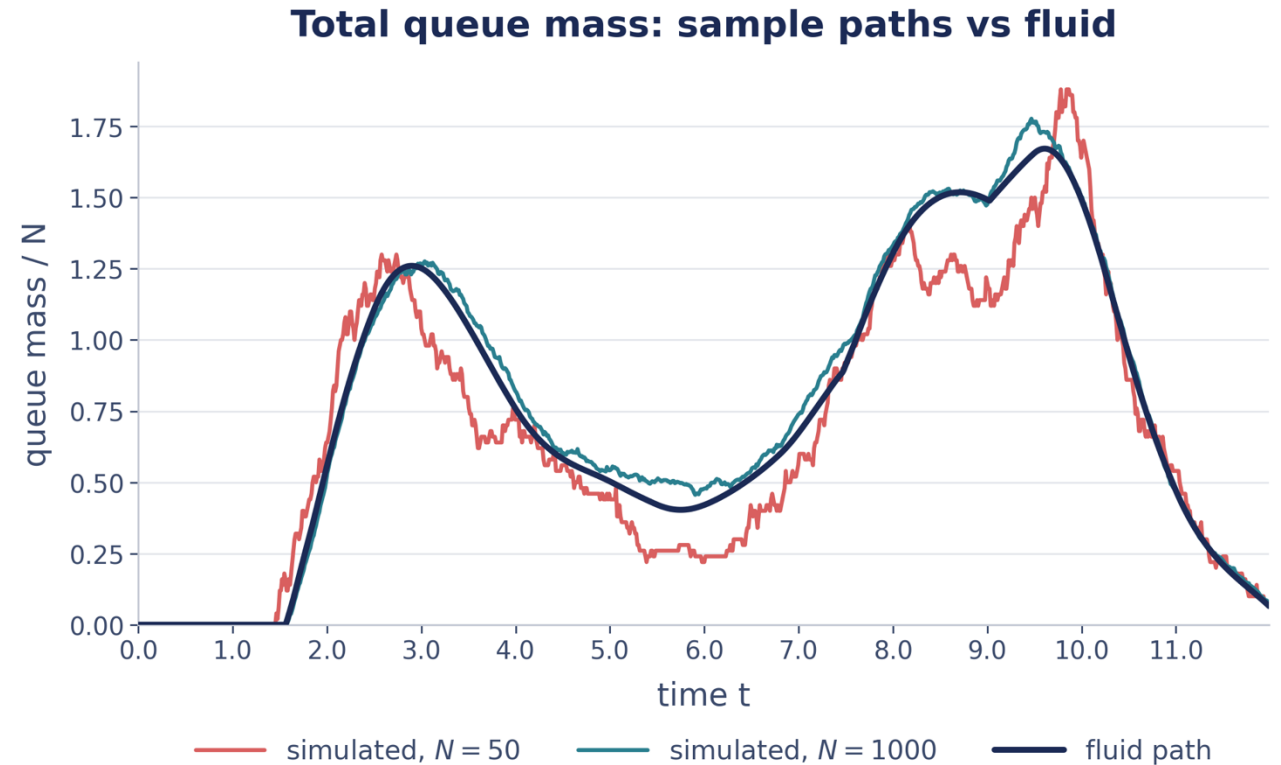
Three-class restaurant day (10 am – 10 pm)

cost per unit scale $C/N \cdot N = 800 \cdot 100 \text{ seeds} \cdot \text{whiskers } \pm 1 \text{ std}$



Pass to the Many-Server Fluid Limit

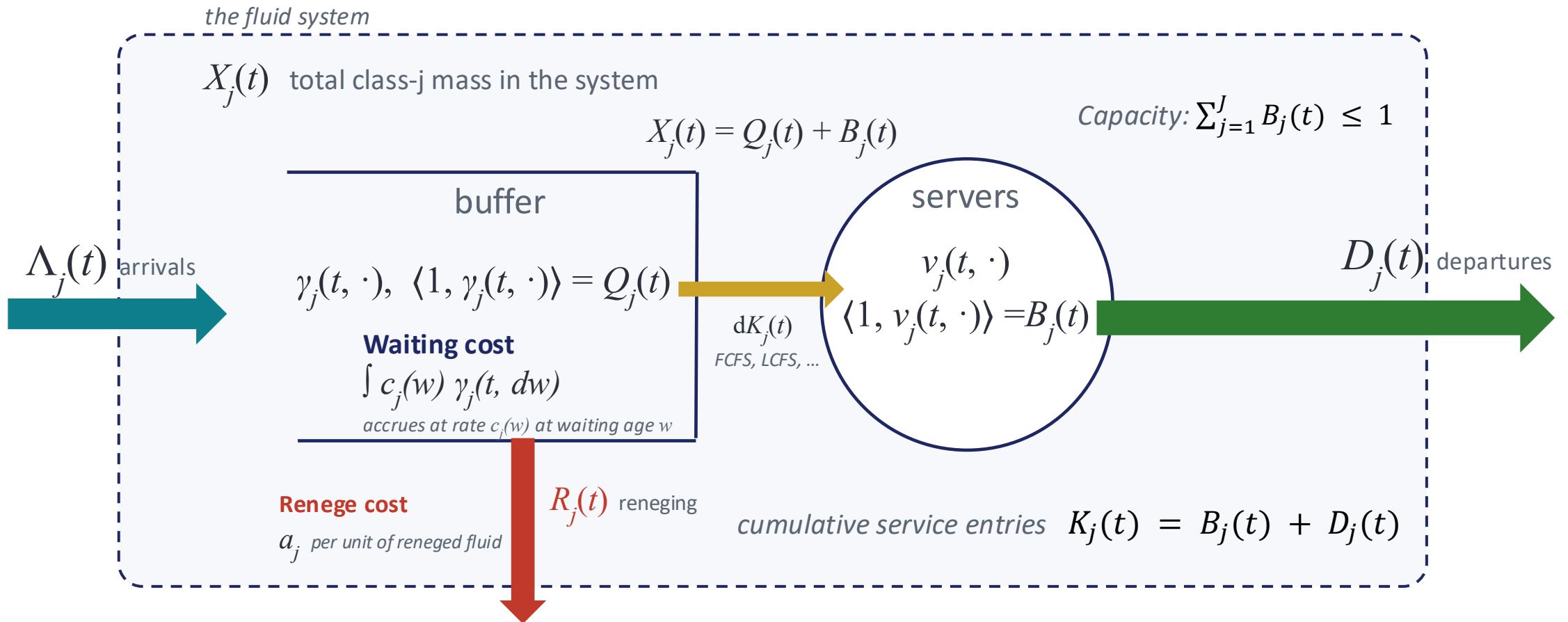
- Scale the system: **arrival rates** and the server count both grow proportionally with N — over the fixed horizon $[0, T]$
- All queue quantities are normalized by N
- By FSLLN, as $N \rightarrow \infty$, the G/GI/ N +GI system concentrates on a Cadlag deterministic path — the fluid limit; $(\gamma_j^N, v_j^N) / N \rightarrow (\gamma_j, v_j)$ as $N \rightarrow \infty$



Selected Literature:

Literature	Model	Scale	Horizon	Policy Class	Proposed Policy	Cost
Smith (1961), Büyükkoç (1985), — $c\mu$ rule	multiclass G/M/1	-	Finite	Preemptive	$c\mu$ -rule	linear holding
Mieghem (1995), Mandelbaum–Stolyar (2004) — $Gc\mu$ rule	multiclass G/G/1	heavy-traffic	Finite	Preemptive	$Gc\mu$ -rule	convex waiting
Whitt (2006), Kang–Ramanan (2010), Liu–Whitt (2012)	1-class G/GI/N+GI time-varying arrivals	fluid	finite & ∞	—	FCFS	— (limit theory)
Atar–Kaspi–Shimkin (2014), Puha–Ward (2022)	multiclass time-varying arrival G/GI/N+GI	fluid	finite & ∞	—, HL	Static priority, $c\mu/\theta$, WRBS	$\sim$ limit theory
Atar–Giat–Shimkin (2010) — $c\mu/\theta$	multiclass M/M/N+M	fluid	∞ , average	Preemptive	$c\mu/\theta$ -rule	linear holding
Bassamboo–Randhawa–Wu (2023)	multiclass GI/GI/N+GI	fluid	∞ , average	Age-based policy	mTIQ	age-dep. cost
Long et al. (2020)	multiclass G/M/N+GI	fluid	∞ , average	HL, work-conserving	Target-allocation, $Gc\mu/h$	holding + renege cost
Dong–Ibrahim (2021)	1-class M/GI/N+GI	fluid	∞ , average	preemptive, sizes known	SRPT	$\sim$ abandonment cost
This paper	multiclass G/GI/N+GI, time-varying arrivals	fluid	finite [0,T]	non-HL, non-preemptive, idling allowed	Tagged	age-dep. + renege cost

The Fluid Limit of Our Problem



Mass balance: $X_j(t) = \Lambda_j(t) - R_j(t) - D_j(t)$

Fluid Model Solution: The Formal Definition

Fluid model solution

Cadlag (X, ν, γ) is a fluid model solution if, for every class j , and $t \in [0, T]$:

(i) Service age: $\langle f, \nu_j(t, \cdot) \rangle = \int_0^t f(t-u)(1 - G_j^s(t-u)) dK_j(u)$

(ii) Waiting age: $\langle 1, \gamma_j(t, \cdot) \rangle = Q_j(t)$, with renewal bound

$$\gamma'_j(t, A) - \gamma'_j(s, A) \leq \lambda_j(A \cap (s, t]) \quad \forall A \in \mathcal{B}([0, T])$$

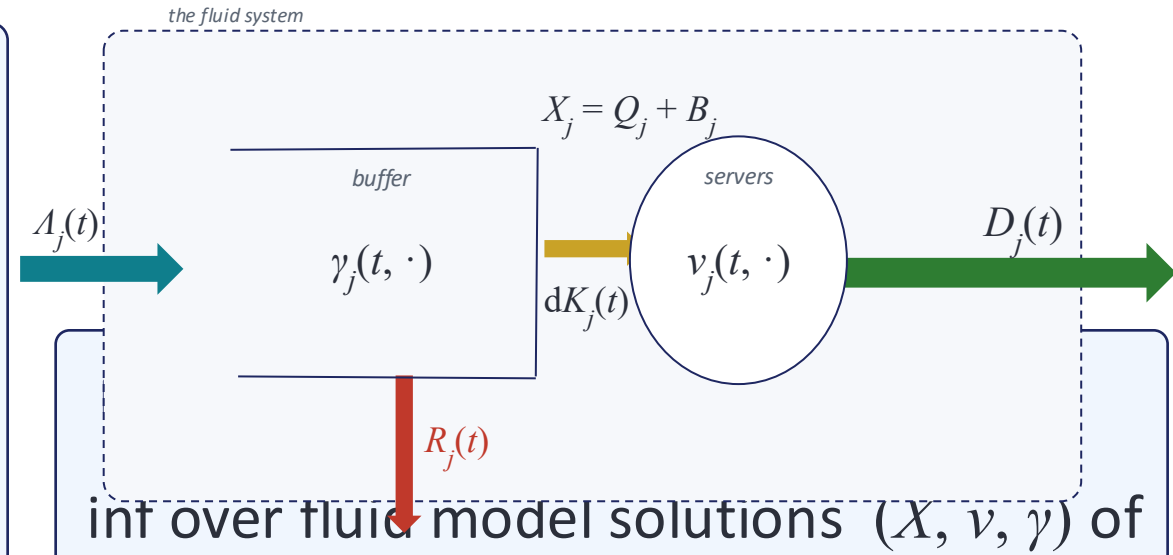
where $\gamma'_j(t, A) = \int_{[0, t] \cap [0, H_j^r)} 1_A(t-w) \frac{\gamma_j(t, dw)}{s_j^r(w)}$

(iii) Mass balance: $X_j(t) = \Lambda_j(t) - R_j(t) - D_j(t)$,

$$R_j(t) = \int_0^t \langle h_j^r, \gamma_j(u, \cdot) \rangle du, \quad D_j(t) = \int_0^t \langle h_j^s, \nu_j(u, \cdot) \rangle du$$

(iv) Capacity: $\sum_{j=1}^J B_j(t) \leq 1$

(v) Monotonicity: K_j is non-decreasing



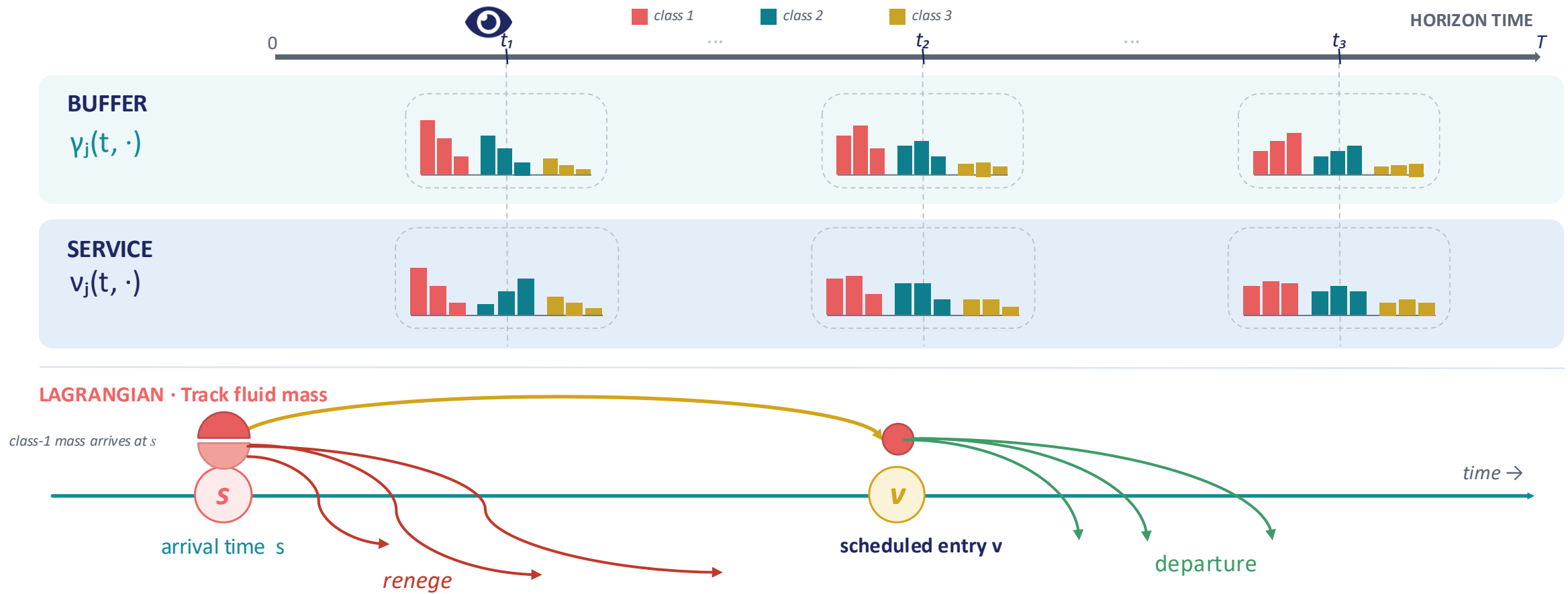
Theorem (bijection)

fluid controls $k = (k_j)_j \Leftrightarrow$ fluid model solutions (X, ν, γ)

k_j a Radon measure on $[0, T] \times [0, H_j^r)$,

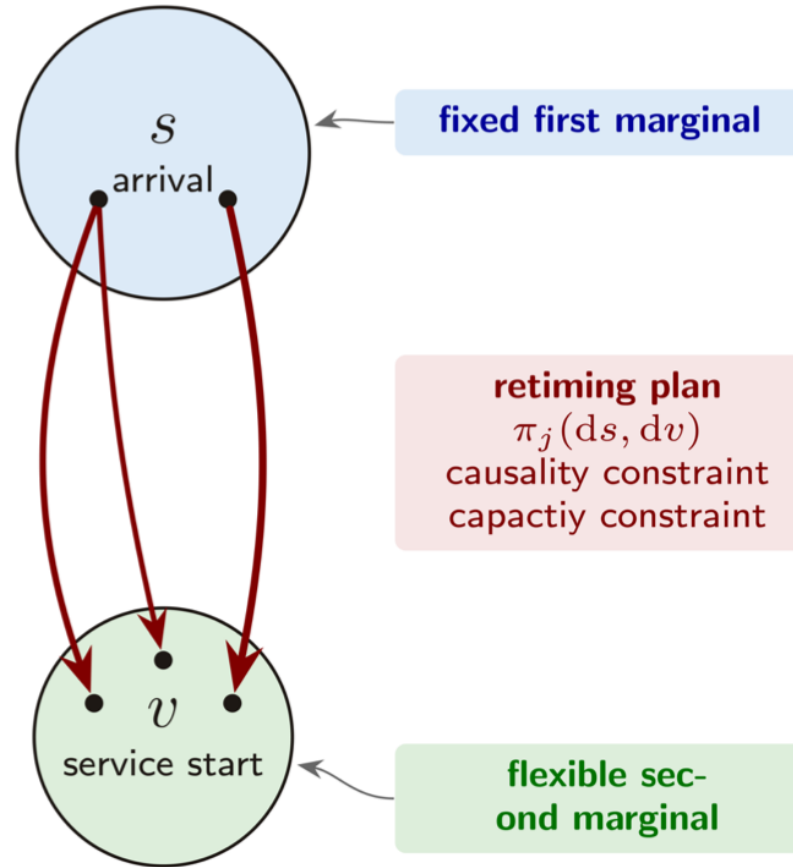
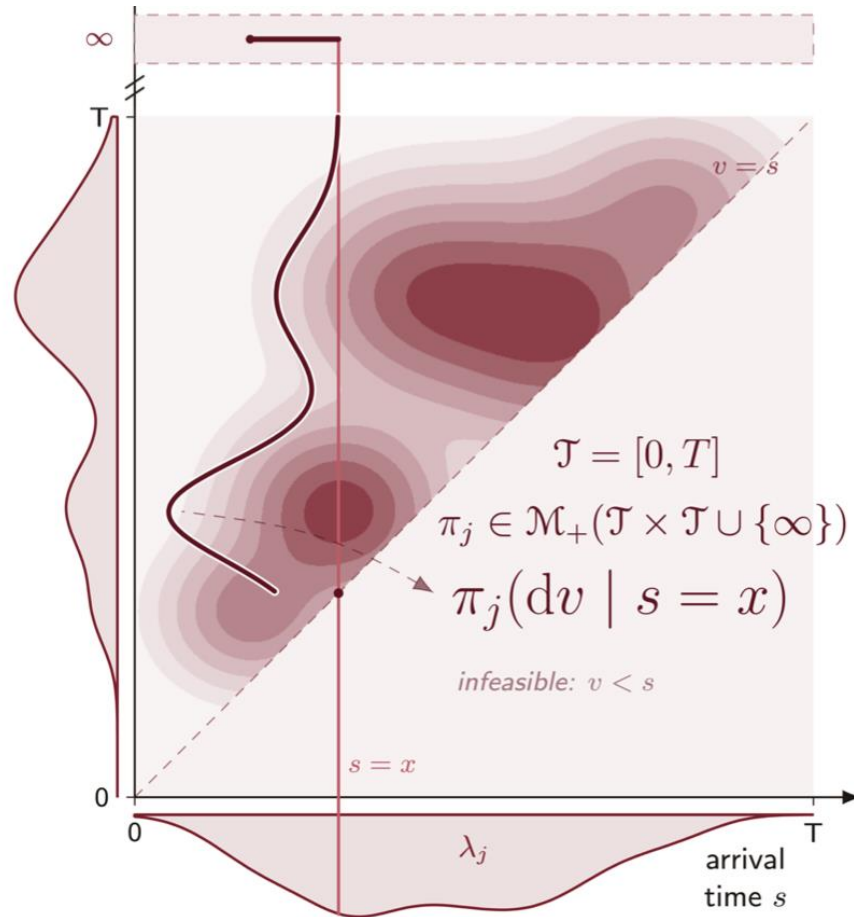
$$dK_j(t) = k_j(t, [0, H_j^r))$$

Eulerian and Lagrangian Fluid Coordinates



states $(\gamma_j(t, \cdot), v_j(t, \cdot)) \leftrightarrow$ particle events (s, v)

Retiming Plan: One Static Decision Object



Reformulation As LP

Feasible retiming plans $\Pi(D)$

$\pi_j \in \mathcal{M}_+(T \times T \cup \{\infty\})$, $T = [0, T]$, s arrival, v intended start

(i) fixed arrival marginal: $\pi_j(A \times T \cup \{\infty\}) = \lambda_j(A)$

(ii) causality: $\text{supp}(\pi_j) \subseteq \{(s, v) : 0 \leq s \leq v \leq T\} \cup \{T \times \{\infty\}\}$

(iii) capacity, for all t :

$$\sum_j \iint_{s \leq v \leq t} S_j^s(t - v) S_j^r(v - s) \pi_j(ds, dv) \leq 1$$

The (RO) linear program

cost of one assignment (s, v) :

$$d_j(s, v) = \int_0^{v-s} c_j(w) S_j^r(w) dw + a_j G_j^r(v - s)$$

$$(RO): \inf_{\pi \in \Pi(D)} \sum_j \int d_j(s, v) \pi_j(ds, dv)$$

- one **static LP** over measures
- the whole schedule is a single object π — no dynamic program on a measure-valued state space

Theorem (Bijection). Feasible retiming plans $\pi \iff$ fluid model solutions (X, v, γ)

Duality: Prices for Tables and Arrivals

Dual Program

$$\begin{aligned}
 & \sup_{y, \varphi} \sum_{j=1}^J \int_{\mathcal{T}} \varphi_j(s) \lambda_j(ds) - \int_{\mathcal{T}} y(v) dv \\
 & \text{s.t. } y \in \mathcal{L}_+^\infty(\mathcal{T}), \quad \varphi_j \in \mathcal{L}^1(\Lambda_j), \quad \varphi_j(s) \leq c_j(s, \infty) \\
 & \quad \varphi_j(s) \leq d_j(s, v) + S_j^s(v-s) (S_j^{r\uparrow} y)(v), \quad s \leq v. \\
 & \quad p_j(v) = (S_j^{r\uparrow} y)(v) = \int_v^T S_j^r(u \uparrow v) y(u) du.
 \end{aligned}$$

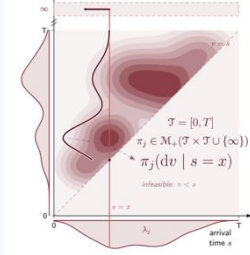
arrival potential value $\varphi_j^*(s)$

capacity shadow price $y^*(t)$

future congestion charge $p_j^*(v)$

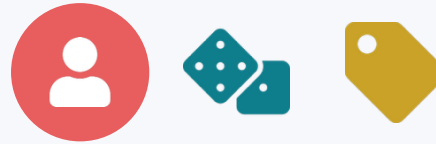
The Proposed Policy: Tagged

1 SOLVE (RO) ONCE — OFFLINE



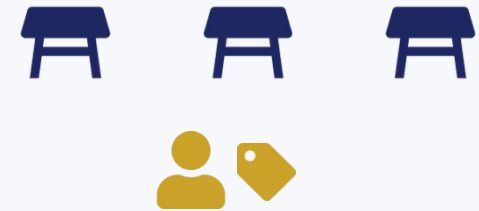
- Given the instance:
 $(\lambda_j, \text{service}, \text{patience}, c_j, a_j, N, T)$
- Solve the static LP once — optimal plan π^*
- Read off the start kernel:
 $\pi_j^*(ds, dv) = \lambda_j(ds) \kappa_j^*(s, dv)$

2 TAG EACH ARRIVAL



- A class- j customer arrives at time s
- Draw a target start time $v \sim \kappa_j^*(s, \cdot)$
- Attach the tag (no re-solving)

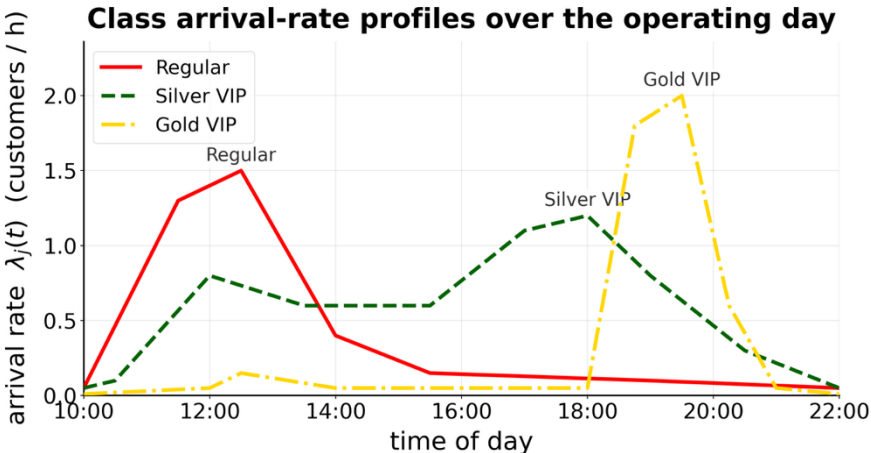
3 SEAT BY TARGET TIME



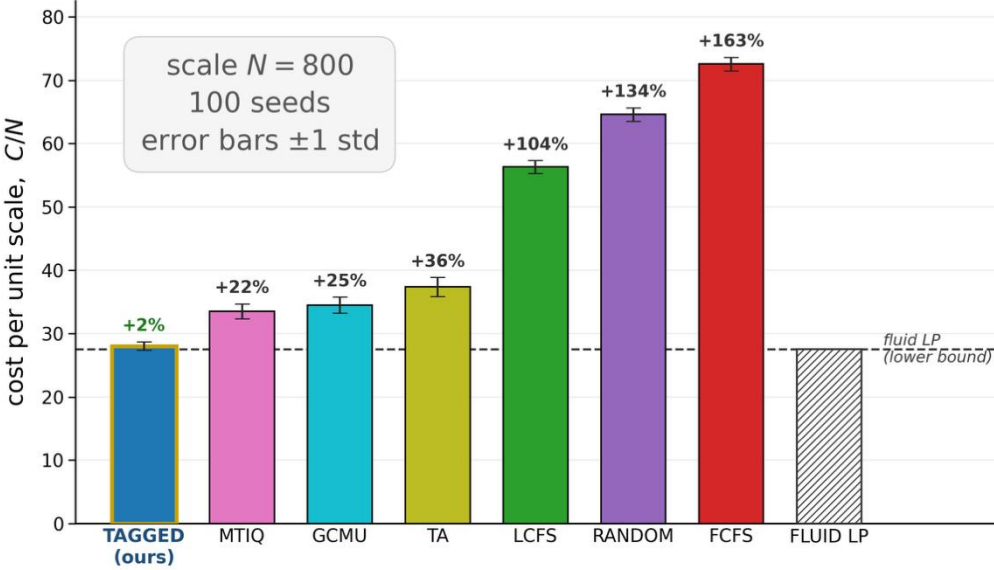
- Table available: admit the waiting customer with the earliest target time among all
- Customers still renege if patience runs out first
- Idle if the earliest target time < clock

Numerical Results : Recapture

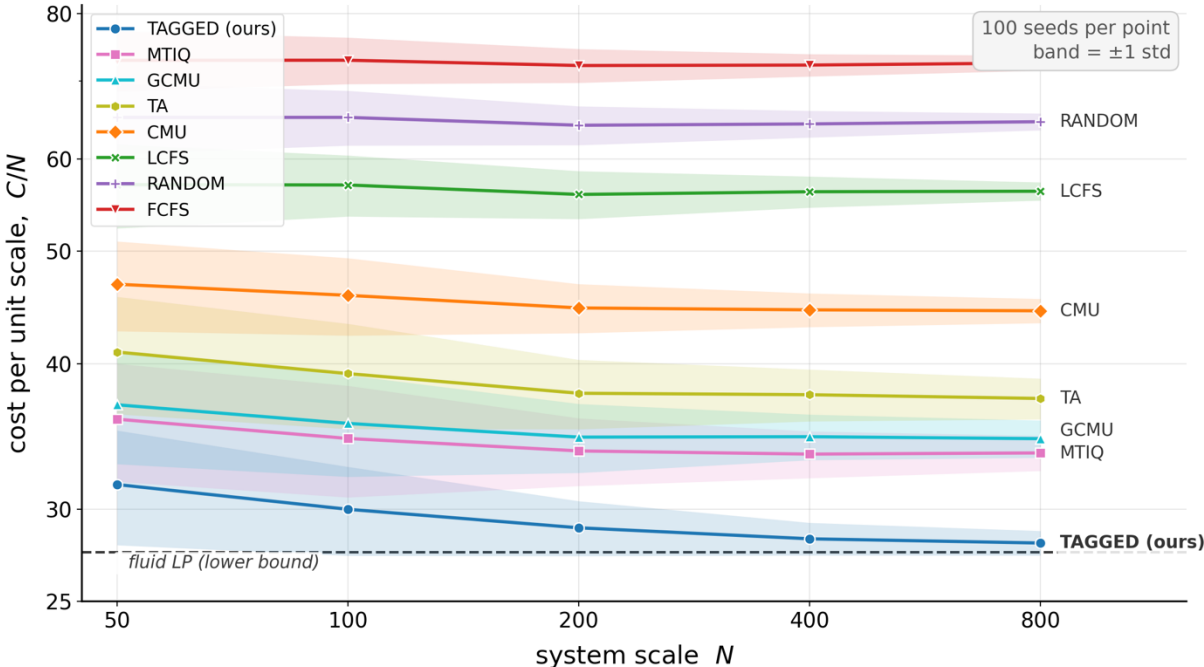
Property	Class 1 — Regulars	Class 2 — Silver VIPs	Class 3 — Gold VIPs
Arrivals	lunch peak (avg 0.41/h)	all day (avg 0.62/h)	dinner wave (avg 0.32/h)
Patience	Medium (~0.67 h)	High (~1.5 h)	Low (~0.5 h)
Service	Fast (~0.67 h)	Slow (~2.0 h)	Medium (~0.83 h)
Abandonment cost	Low (1)	Medium (3)	High (15)
Holding cost	Low ($0.5 \cdot w^2$)	Medium ($1.0 \cdot w^2$)	High ($2.0 \cdot w^2$)



Three-class restaurant day (10am-10pm)



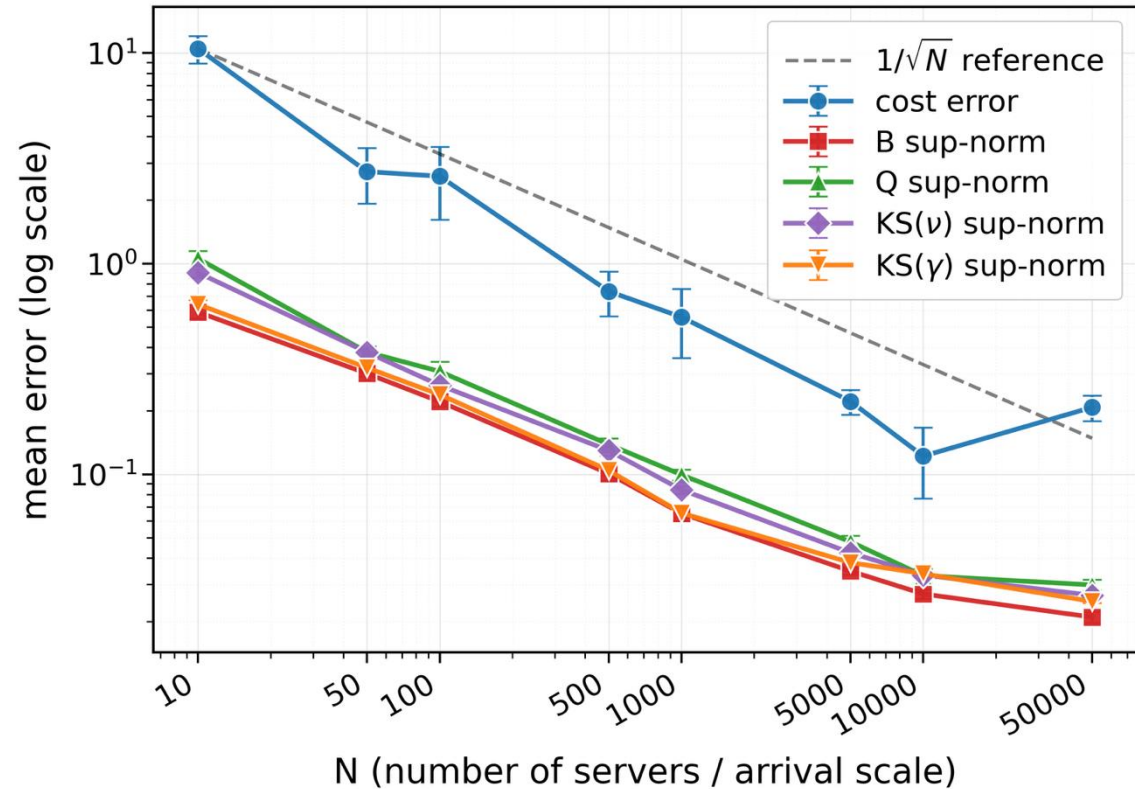
Three-class restaurant day (10am-10pm): policy cost vs. system scale



Numerical Convergence Check

- Question: Does the scaled stochastic system really converge the fluid model as N grows?

Restaurant instance: convergence errors vs N (10 reps, mean $\pm$ SE)



Takeaways and What's Next

- *Fluid scheduling of multiclass $G/GI/N+GI \Rightarrow$ equivalent static LP over measures (Retiming Optimization)*
- *Implementable tagged-start-time policy, validated numerically*
- *Extensions: robustness, multiclass servers, queueing networks, dependency*

Thank you!

Questions welcome · haoyuwu0@chicagobooth.edu